Retrace: Frontend Design and Workflow Specification

Version: 1.0 | **Date:** 10 November 2026 | **Status:** Implementation baseline for review **Applies to:** `web/` (React SPA), its generated API types, and every coding-agent task that touches the frontend **Visual direction:** Hybrid. Dense, calm dashboard shell around an editorial, writing-first postmortem editor. **Dark theme only in v1.**

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0. How to use this document

This is one document with two jobs: (A) the **design** (how Retrace looks, what components exist, how the app is structured) and (B) the **workflow** (every screen, every control, where it leads, what it calls, what happens on success and failure). It is written so an implementation plan can be cut directly from it, one chapter or one screen at a time.

0.1 Reading order for humans and agents

1. Part 1 to 3: product frame, design system, app architecture. Read once, before any task.
2. Part 4: the screen index. Every screen has a stable ID (`S-01` to `S-69` matching the master inventory) and every module a stable ID (`M01` to `M21`).
3. Part 5: cross-cutting behavior. Every screen inherits it; module chapters do not repeat it.
4. Part 6: module chapters. Each one is self-contained and is the unit of an implementation task.
5. Part 7: end-to-end workflows and state machines.
6. Part 8: build plan, task template, definition of done.
7. Part 9: gaps and assumptions register. Read before touching anything marked **GAP**.

0.2 Stable identifiers (use these in task prompts, commits and PRs)

Prefix	Meaning	Example
`S-nn`	Screen or surface from the master inventory	`S-19` Postmortem editor
`Mnn`	Module chapter	`M10` Postmortem editor
`Mnn-Xm`	Interaction row in a module	`M12-X04` Approve button
`Mnn-ACk`	Acceptance criterion	`M10-AC07`
`WF-n`	End-to-end workflow	`WF-2` Incident to approved postmortem
`SM-n`	State machine	`SM-1` Postmortem
`GAP-n`	Missing contract detail that blocks or constrains frontend work	`GAP-04`

0.3 Authority and labels

This document must not contradict the PRD v0.6, System Design, Database Design v0.1, AI Pipeline Spec v0.2, API and Event Contracts v1.0, or Repository Conventions v1.0. Where they disagree, use the reconciliation order in Repository Conventions section 1. Labels used here:

Label	Meaning
Fixed	Decided in a source document. Implement as written.
Design choice	Decided in this document. Visual and interaction decisions only; changeable by editing this file.
GAP	A source document leaves the detail open. Do not invent it. Build against the documented fallback and record the gap in the PR.
P0 / P1 / P2 / Later	Priority from the PRD and master inventory.

0.4 Non-negotiable frontend rules

1. **No invented endpoints, fields, numeric limits or provider behavior.** Only the endpoint inventory in API Contracts section 4 exists. Anything else is a GAP.
2. **UI hiding is cosmetic.** Every action also fails safely when the server returns 403 or 404.
3. **Every claim in a postmortem stays traceable.** A citation chip always resolves to a timeline event.
4. **AI output is a suggestion until a human accepts it.** The UI never writes AI text into a document without an explicit accept action.
5. **Facts, hypotheses and open questions are always visually distinct.** Never rely on color alone.
6. **A stored secret is never rendered back.** Only a masked hint.

7. **Tenant context is explicit.** Every tenant request carries the active `org_id` from the route.
8. **Never display "verified" when semantic verification was unavailable.**
9. **Generated API types in `web/src/generated/` are never hand-edited.**

PART 1. PRODUCT FRAME FOR THE FRONTEND

1.1 What the user is trying to do

Retrace turns a resolved incident into a reviewed, blameless postmortem and tracked follow-up work. The frontend exists to make one loop feel short and trustworthy:

Describe, Discover, Confirm evidence, Generate, Edit, Blameless check, Review, Approve, Publish, Track actions.

The only required author input per incident is one short description. Everything else is discovered, inferred, or optional. Every screen decision below is judged against that.

1.2 Experience principles (Design choices)

Principle	What it means on screen
Evidence first	Source, time and a link back to the original are always one click away from any statement.
Quiet by default	Dense data lives in the shell. The editor is calm: wide margins, serif body, no chrome until needed.
Honest uncertainty	Hypotheses, open questions, discrepancies and "verification unavailable" are shown plainly, never hidden.
Human decides	Every AI action ends in Accept or Reject. Every approval is an explicit button.
Never lose work	Autosave with visible status, conflict detection, retry on failed jobs, no silent overwrite.
Progress is visible	Every background job shows state, progress and a user-safe error with a retry.

1.3 Roles in the UI

v1 builds **Admin** and **Member**. Manager and Viewer arrive later, so the UI is built around a **capability map**, not hard-coded role names (see 3.7). Adding a role later means editing one table.

Role	v1	UI summary
Admin	Built	Everything, including members, integrations, AI model, audit log, approve, publish.
Member	Built	Create incidents, generate drafts, edit, comment, manage action items, view analytics, export.
Manager	Later	Manage catalog, approve. Until enabled, Admins approve.
Viewer	Later	Read published postmortems and analytics; comment only.

1.4 Scope by phase (matches the 8-week plan)

Phase	Weeks	Frontend deliverables
1. Foundation	1 to 2	Shell, auth screens, org creation and "setting up" screen, org switcher, invitations, members, incident list/create/detail shell.
2. Core loop	3 to 5	Integrations, allowlists, service catalog, evidence preview, timeline, generation progress, editor with inline AI, citations and comments, blameless panel, review/approval, versions, PDF/Markdown export. Stretch: GitHub screens.
3. Close the loop	6 to 8	Action items and reminders UI, notifications, analytics (recovery time, volume, action health), audit log, AI usage, polish, demo data views.
Later	After v1	Real-time co-editing, repeat-incident UI, audit-ready reports, Manager/Viewer, teams, 2FA, org archive, share links, offline, voice.

PART 2. DESIGN SYSTEM (DARK ONLY)

All values are **Design choices**. They are expressed as CSS variables so a light theme can be added later without touching components. Set `color-scheme: dark` on `:root`.

2.1 Principles

- Dark surfaces layered by elevation, not by heavy borders.
- One accent color. Status and claim colors carry meaning; the accent carries action.
- Minimum body contrast 4.5:1, large text 3:1, focus rings always visible.
- Color is never the only signal: pair with an icon or text label.

2.2 Color tokens

Token	Value	Use
`--bg`	`#0B0D10`	App background
`--surface-1`	`#111418`	Cards, panels, sidebar
`--surface-2`	`#171B21`	Raised: popovers, menus, table header
`--surface-3`	`#1E232B`	Hover, selected row
`--border`	`#232932`	Default hairline
`--border-strong`	`#2F3744`	Inputs, focused containers
`--text`	`#E6E9EE`	Primary text
`--text-muted`	`#9AA3B0`	Secondary text
`--text-faint`	`#6B7480`	Placeholder, disabled
`--accent`	`#7C8CFF`	Primary buttons, links, focus ring, active nav
`--accent-hover`	`#95A2FF`	Hover
`--accent-soft`	`rgba(124,140,255,.14)`	Selected backgrounds
`--danger`	`#F47067`	Destructive, errors
`--warning`	`#E3B341`	Warnings, stalled
`--success`	`#56D364`	Success, done
`--info`	`#58A6FF`	Informational

Severity (incident)

Severity	Token	Color	Label
sev1	`--sev1`	`#F47067`	SEV1 (critical)
sev2	`--sev2`	`#F0883E`	SEV2 (major)
sev3	`--sev3`	`#E3B341`	SEV3 (minor)
sev4	`--sev4`	`#8B949E`	SEV4 (low)

AI claim kinds (postmortem text and citation UI)

Each kind has a color, an icon and a text label. Hypotheses must never look like facts.

Claim kind	Color	Icon	Text treatment in editor
`observed_fact`	`#56D364`	check-circle	Normal text, solid citation chips
`causal_claim`	`#58A6FF`	link	Normal text, chips plus a "causal" badge on hover
`hypothesis`	`#E3B341`	flask	Dashed underline, "Hypothesis" tag, amber chip
`open_question`	`#B392F0`	help-circle	Italic, "Open question" tag
`author_context`	`#76E3EA`	user	Cyan left rule, "Author context, not verified" tag
`process_recommendation`	`#9AA3B0`	lightbulb	Muted, "Proposal" tag

2.3 Typography

Self-host fonts through `@fontsource` packages (no external font request).

Role	Family	Fallback	Use
UI	Inter	system-ui, sans-serif	All shell UI
Editorial	Source Serif 4	Georgia, serif	Postmortem editor body and exported-view reading mode
Mono	JetBrains Mono	ui-monospace, monospace	IDs, timestamps, code, evidence snippets

Style	Size / line	Weight
Page title	24 / 32	600
Section title	18 / 26	600
Body (UI)	14 / 22	400
Small / meta	12 / 18	400
Editor body	17 / 28	400
Editor H1 / H2	30 / 38, 22 / 32	600

2.4 Spacing, radius, elevation, layout

- 4px base unit. Scale: 4, 8, 12, 16, 24, 32, 48, 64.
- Radius: 6 (inputs, chips), 10 (cards), 14 (dialogs). Full radius for avatars and pills.
- Elevation by surface step and a subtle border; shadow only on popovers and dialogs.
- Shell: left sidebar 240px (collapsible to 64px icon rail), top bar 56px, content max-width 1280px for dashboards.
- **Editor layout:** centered column max 720px, generous vertical rhythm, side panels slide over from the right (360 to 420px) and never reflow the column.

- Breakpoints: laptop 1024+ is the design target; tablet 768 to 1023 collapses sidebar to icon rail and side panels become overlays. Below 768 is out of scope (PRD: laptop and tablet).

2.5 Motion

150ms ease-out for hover and focus, 200ms for panel slide and dialog, 300ms max for anything else. Respect `prefers-reduced-motion` (remove sliding, keep opacity). Streaming AI text appears incrementally with no typing-animation flourish.

2.6 Accessibility (Fixed by PRD usability requirement, details are Design choices)

Keyboard navigation for all core flows, visible focus ring (2px accent, 2px offset), ARIA labels on icon-only buttons, dialogs trap focus and restore it on close, toasts use `aria-live="polite"`, errors use `role="alert"`, tables are real tables with headers, status is never color-only.

2.7 Component library

Built on shadcn/ui primitives, themed with the tokens above. Every component below is a shared component in `src/components/` unless it is feature-specific (then it lives in its feature folder).

Group	Components
Primitives	Button (primary, secondary, ghost, danger, icon), Input, Textarea, Select, Combobox, Checkbox, Switch, RadioGroup, Tabs, Tooltip, Popover, DropdownMenu, Dialog, AlertDialog, Sheet (side panel), Toast, Skeleton, Badge, Avatar, Kbd
Data	DataTable (sort, filter row, cursor/page pagination), Pagination (page), CursorPager, FilterBar, SortMenu, EmptyState, ErrorState, PermissionDenied, StatCard, TrendChart (Recharts)
Domain	SeverityBadge, StatusBadge (incident, postmortem, action item, evidence, job), RoleBadge, SourceIcon (slack, github, pagerduty, manual), ClaimKindTag, CitationChip, TimestampLabel (with precision and timezone-unknown marker), DiscrepancyFlag, ConfidenceMeter, UserChip, ServiceChip
Workflow	JobProgress, StaleVersionDialog, ConfirmDestructiveDialog, ApprovalBar, SaveStatus, VerificationStatusBanner, LimitationsList
Editor	PostmortemEditor (TipTap), CitationNode, CommentMark, AIActionMenu, SuggestionDiff, SectionNav, BlamelessPanel, CommentPanel, CitationPanel
Shell	AppShell, Sidebar, TopBar, OrgSwitcher, NotificationBell, UserMenu, Breadcrumbs, SettingUpScreen

2.7.1 Key component contracts

StatusBadge takes `kind` (`incident|postmortem|action|evidence|job|generation`) and `value`, and maps to label, color and icon in one lookup table. No screen builds its own badge.

TimestampLabel takes the five evidence timestamp fields (`occurred_at`, `occurred_at_raw`, `observed_at`, `timestamp_precision`, `timezone_known`). It renders:

- UTC value in mono when precise.
- Precision hint (for example "minute precision") when not exact.
- A warning icon with tooltip "Timezone not known" when `timezone_known` is false.
- It never formats a date-only value as an exact instant.

CitationChip takes a claim id, one or more timeline event ids and a claim kind. Click opens the CitationPanel on the first event; chips with several events show a count and open a list.

JobProgress takes a `job_id`, polls `GET /jobs/{job_id}` (start 2s, back off), later upgrades to SSE (GAP-03). It renders queued/running/succeeded/failed/cancelled, progress percent as an indeterminate-friendly bar (never infer hidden AI stages from the integer), user-safe error, and a Retry button where retry is allowed.

ConfirmDestructiveDialog requires an explicit confirm button labeled with the action ("Delete incident"). For org deletion and incident deletion, also requires typing the resource name.

SaveStatus states: `Saving...`, `Saved`, `Offline or failed (Retry)`, `Conflict (Reload)`.

PART 3. APPLICATION ARCHITECTURE

3.1 Stack (Fixed)

React + Vite + TypeScript, Tailwind CSS, shadcn/ui, React Router, TanStack Query, TipTap (ProseMirror JSON), typed client generated from OpenAPI, pnpm, ESLint + Prettier, Vitest + React Testing Library + Playwright. Hosted on Azure Static Web Apps, calling the gateway on the `api` subdomain.

Additional libraries (Design choices, all optional to swap): Recharts (charts), React Hook Form + Zod (forms), `diff` or `jsdiff` (version compare), `date-fns` (dates, UTC display), `lucide-react` (icons), `cmdk` (command palette and search, P1).

3.2 Folder structure (Fixed shape, Design-choice contents)

```
web/src/
    app/                        # bootstrap, router, providers, guards
      router.tsx
      providers.tsx             # QueryClient, Auth, Org, Toast, Theme
      guards/                   # RequireAuth, RequireOrg, RequireCapability
    features/
      auth/                     # M01
      orgs/                     # M02 (orgs, members, invitations, switcher)
      dashboard/                # M03
      incidents/                # M04
      integrations/             # M05
      services/                 # M06
      evidence/                 # M07
      timeline/                 # M08
      generation/               # M09
      postmortem/               # M10, M11, M12, M13 (editor, blameless, review, version
  s)
      actions/                  # M14
      analytics/                # M15
      search/                   # M16
      exports/                  # M17
      audit/                    # M18
      notifications/            # M19
      ai-settings/              # M20
    components/                 # shared presentational components (2.7)
    api/                        # handwritten query/mutation functions + client wiring
      client.ts                 # fetch wrapper: auth, org, ETag, Idempotency-Key, errors
      queryKeys.ts
      <service>/*.ts            # core, ingestion, ai, notifications
    generated/                  # committed generated types (never hand-edit)
    hooks/                      # useJob, useAutosave, useCapability, useOrg, useDebounce
    lib/                        # dates, ids, errors, permissions, format
    styles/                     # tokens.css, globals
    test/                       # fixtures, MSW handlers, render helpers
```

Rule: code is promoted from a feature to `components/` or `hooks/` only when a genuine second consumer exists.

3.3 Route map

All authenticated, tenant routes live under `/o/:org`. `:org` is the organization id (UUIDv7) or its slug resolved to the id by the org context (**Design choice:** use the slug in the URL, keep the id in context; every API call uses the id).

Route	Screen(s)	Guard
`/login`, `/signup`	S-01, S-02	Public (redirect if signed in)
`/verify-email`	S-03	Public (token in query)
`/forgot-password`, `/reset-password`	S-04, S-05	Public
`/auth/callback/:provider`	S-06	Public (OAuth return)
`/accept-invite`	Invitation acceptance	Public, then auth
`/orgs/new`	S-08	Auth
`/o/:org`	S-09 Dashboard	Auth + org
`/o/:org/incidents`	S-10	Auth + org
`/o/:org/incidents/new`	S-11 (modal route over list)	Auth + org, can `incident.create`
`/o/:org/incidents/:id`	S-12 Overview tab	Auth + org
`/o/:org/incidents/:id/evidence`	S-13, S-14	Auth + org
`/o/:org/incidents/:id/timeline`	S-15, S-16	Auth + org
`/o/:org/incidents/:id/postmortem`	S-18 to S-23, S-26, S-27	Auth + org
`/o/:org/incidents/:id/postmortem/versions`	S-24	Auth + org
`/o/:org/incidents/:id/postmortem/versions/compare?a=&b=`	S-25	Auth + org
`/o/:org/incidents/:id/actions`	Incident action items	Auth + org
`/o/:org/incidents/:id/activity`	Incident activity	Auth + org
`/o/:org/postmortems`	S-17 Library	Auth + org
`/o/:org/actions`	S-29	Auth + org
`/o/:org/actions/:itemId`	S-31 (drawer route)	Auth + org
`/o/:org/analytics` (+ `/recovery`, `/volume`, `/action-health`)	S-33 to S-36	Auth + org
`/o/:org/notifications`	S-46	Auth + org
`/o/:org/account/profile`	S-07	Auth
`/o/:org/account/notifications`	S-47	Auth
`/o/:org/settings/general`	S-42	Admin
`/o/:org/settings/members` (+ invitations tab)	S-43, S-44	Admin
`/o/:org/settings/integrations` (+ `/:id`, `/:id/allowlist`)	S-37 to S-39	Admin

Route	Screen(s)	Guard
`/o/:org/settings/services`	S-40, S-41	Admin write, members read (see M06)
`/o/:org/settings/ai-model`	S-45	Admin
`/o/:org/settings/audit-log` (+ `/verify`)	S-48, S-49	Admin
`/o/:org/search`	S-52 (P1)	Auth + org

Deviation note: the System Design lists notification settings under `settings/*` (Admin). Because notification preferences are per user (`GET/PATCH /notification-settings`), this document places them under `account/` so Members can reach them. Flagged as **GAP-12** for confirmation.
Unknown tenant resource id returns the not-found screen (never a "forbidden" that discloses existence). An `:org` the user is not a member of renders the same not-found screen.

3.4 API client layer

One fetch wrapper in `api/client.ts`. Handwritten functions per endpoint call it; generated types provide request/response shapes.

Concern	Behavior
Base URL	Gateway at the `api` subdomain, path prefix `/api/v1`. Tenant calls: `/api/v1/orgs/{org_id}/...`.
Auth	`Authorization: Bearer <access token>` held in memory only. Refresh token is an HttpOnly cookie; send `credentials: "include"` to `/auth/refresh`.
Silent refresh	On 401, call `POST /auth/refresh` once (single-flight: concurrent 401s share one refresh), replay the request. If refresh fails, clear session and go to `/login?next=...`.
Org binding	Active org comes from the route `:org`. The access token carries the active org; switching org requires a token for that org (see M02, GAP-11).
ETag	Store the `ETag` response header with the cached resource. Send `If-Match` on PUT/PATCH/DELETE for mutable important resources. Never construct an ETag.
Idempotency	For create/command POSTs, generate one UUID per user intent and send it as `Idempotency-Key`; reuse the same key on automatic retry of that intent; new key for a new intent.
Errors	Parse the standard error shape `{ code, message, details?, correlation_id }` into a typed `ApiError`. Map `code` to user text in `lib/errors.ts`. Always keep `correlation_id` for the "Copy reference" affordance.
202 jobs	Commands returning `202 { job_id, status }` hand off to `useJob(job_id)`.
429	Read `Retry-After`; show a toast with the wait and disable the triggering control until then.
Pagination	Cursor lists (timeline, audit, activity) use `useInfiniteQuery` with `cursor` + `limit`. Page lists (incidents, actions, services, notifications, library) use `page` + `page_size`. Page-size defaults are a **GAP** (CFG-01): read from a single frontend config constant until resolved.
Filter and sort	Query params named from resource fields (`status`, `severity`, `service_id`, `owner_user_id`, `created_at_gte`, `created_at_lte`); `sort=-field,field`. Unlisted fields are never sent.

3.4.1 Error code to UI behavior

Status / code	UI behavior
401	Silent refresh, else redirect to login with `next`.
403	Inline PermissionDenied for the action; toast "You do not have permission".
404	NotFound screen for pages; for background fetches, EmptyState.
409	Toast with specific message from `code` (state conflict). Refetch the resource.
412 `stale_version`	StaleVersionDialog (see 5.6).
422	Map `details` field errors onto form fields.
429	Toast with Retry-After countdown; control disabled.
5xx / network	ErrorState with Retry and "Copy reference" (correlation id).

3.5 State management

State type	Where it lives
Server state	TanStack Query. Query keys in `api/queryKeys.ts`, always prefixed by `[orgId, resource, ...]`.
Session and active org	Small React context (`AuthProvider`, `OrgProvider`). Access token in memory only.
URL state	Filters, sort, page, cursor, selected tab, selected item (so views are shareable and back-button safe).
Form state	React Hook Form + Zod, per form.
Editor state	TipTap/ProseMirror document, local until autosaved.
Ephemeral UI	Component state (open panels, dialogs).

Invalidation rules: a mutation invalidates the narrowest affected keys. Generation success invalidates `[orgId, postmortem, incidentId]`, versions, and blameless findings. Evidence include/exclude invalidates evidence list, timeline, and marks the generation as "needs regeneration" (see M07).

3.5.1 Events and the frontend

The browser does **not** consume Redis Streams events. Backend events (for example `draft.ready`, `timeline.ready`, `postmortem.approved`) surface in the UI through job status, SSE, notifications and refetching. This table is the translation layer.

Backend event	How the UI learns of it	UI reaction
`incident.reported`	Incident appears in list on refetch or notification	Incident shows "Resolved, awaiting description" banner
`incident.description_submitted`	Command returned 202 + job	Evidence tab shows discovery JobProgress
`evidence.ready`	Discovery job succeeded	Evidence tab shows preview, awaiting confirmation
`timeline.ready`	Timeline job succeeded after evidence confirm	Timeline tab populates; "Generate postmortem" enabled
`draft.requested`	Command returned 202 + job	Generation progress screen
`draft.ready`	Generation job succeeded; notification	Postmortem tab shows new named version; blameless result shown
`postmortem.submitted`	Command response	Status badge changes; reviewer notification
`postmortem.approved`	Command response; notification	Editor locks (read-only), approval history updates
`tenant.provisioned`	Poll `GET /orgs/{org_id}` status	SettingUpScreen exits to workspace
`incident.deleted`	Command response	Redirect to incident list; related caches dropped
`audit.recorded`	Audit log refetch	None directly

3.6 Auth, session and tenant context

1. On load, call `POST /auth/refresh`. Success restores a session; failure shows login.
2. `GET /me` returns the user and memberships. `OrgProvider` resolves `:org` against memberships; no match renders NotFound.
3. Org switching (OrgSwitcher) requests a token for the chosen org through Core (GAP-11 for the exact call), clears all query caches for the previous org, and navigates to `/o/{newOrg}`.
4. Critical actions (role change, approve, delete) may re-check current role through Core; the UI treats a 403 here as "your permissions changed", refetches `/me`, and shows PermissionDenied.
5. Sign out calls `POST /auth/logout`, clears memory state and all caches, redirects to `/login`.

3.7 Capability map (role gating)

Components ask `can(capability)`; they never check role names. One table in `lib/permissions.ts` maps role to capabilities. Derived from the PRD permission matrix.

Capability	Admin	Member	Manager (later)	Viewer (later)
`members.manage`	Yes	No	No	No
`integrations.manage`, `aiModel.manage`, `audit.view`	Yes	No	No	No
`services.manage`	Yes	No	Yes	No
`services.view`	Yes	Yes	Yes	Yes
`incident.create`, `draft.generate`	Yes	Yes	Yes	No
`postmortem.edit`	Yes	Yes	Yes	No
`postmortem.comment`	Yes	Yes	Yes	Yes
`postmortem.approve`, `postmortem.publish`	Yes	No	Yes (approve)	No
`actions.manage`	Yes	Yes	Yes	No
`analytics.view`, `export.create`, `postmortem.view`	Yes	Yes	Yes	Yes

Disabled-with-reason beats hidden when the user can see the feature but a guard blocks it (for example "Submit" disabled with tooltip "Run the blameless check first"). Hidden is used only when the capability is never available to the role.

PART 4. SCREEN INDEX

Every screen from the master inventory (S-01 to S-69). **Type** says how it is built: Page, Tab (inside a page), Panel (side sheet), Modal, Drawer, or Section. This resolves the inventory's note that the 69 screens are not 69 routes.

Inv #	Screen	Route / surface	Type	Module	Priority	Roles
S-01	Login	`/login`	Page	M01	P0	Public
S-02	Registration	`/signup`	Page	M01	P0	Public
S-03	Email verification	`/verify-email`	Page	M01	P0	Public
S-04	Forgot password	`/forgot-password`	Page	M01	P0	Public
S-05	Reset password	`/reset-password`	Page	M01	P0	Public
S-06	OAuth callback	`/auth/callback/:provider`	Page (transient)	M01	P0	Public
S-07	User profile	`/o/:org/account/profile`	Page	M01	P0	Auth
S-08	Create organization	`/orgs/new`	Page	M02	P0	Auth
S-09	Dashboard	`/o/:org`	Page	M03	P0	All
S-10	Incident list	`/o/:org/incidents`	Page	M04	P0	All
S-11	Create incident	`/o/:org/incidents/new`	Modal	M04	P0	Admin, Member
S-12	Incident overview	`/o/:org/incidents/:id`	Tab	M04	P0	All
S-13	Evidence discovery	`.../:id/evidence`	Tab	M07	P0	All (write: Admin, Member)
S-14	Evidence preview/review	`.../:id/evidence`	Section	M07	P0	Admin, Member
S-15	Timeline	`.../:id/timeline`	Tab	M08	P0	All
S-16	Timeline event details	Timeline	Panel	M08	P0	All
S-17	Postmortem library	`/o/:org/postmortems`	Page	M10	P0	All
S-18	AI generation progress	Postmortem tab	Section	M09	P0	Admin, Member
S-19	Postmortem editor	`.../:id/postmortem`	Tab	M10	P0	All (edit: Admin,

Inv #	Screen	Route / surface	Type	Module	Priority	Roles
						Member)
S-20	Citation side panel	Editor	Panel	M10	P0	All
S-21	AI suggestions panel	Editor	Panel/inline	M10	P0	Admin, Member
S-22	Blameless findings	Editor	Panel	M11	P0	Admin, Member
S-23	Comments panel	Editor	Panel	M10	P0	All
S-24	Version history	`.../postmortem/versions`	Page/Panel	M13	P0	All
S-25	Version comparison	`.../versions/compare`	Page	M13	P0	All
S-26	Review workspace	Postmortem tab (review mode)	Section	M12	P0	All
S-27	Approval screen	Postmortem tab ApprovalBar	Section/Modal	M12	P0	Admin
S-28	Approval history	Postmortem tab	Panel	M12	P0	All
S-29	Action item list	`/o/:org/actions`	Page	M14	P0	All
S-30	Create/edit action item	Modal/Drawer	Modal	M14	P0	Admin, Member
S-31	Action item details	`/o/:org/actions/:itemId`	Drawer	M14	P0	All
S-32	Action item verification	Drawer step	Modal	M14	P0	Owner, Admin
S-33	Analytics overview	`/o/:org/analytics`	Page	M15	P0	All
S-34	Recovery-time analytics	`/o/:org/analytics/recovery`	Tab	M15	P0	All
S-35	Incident-volume analytics	`/o/:org/analytics/volume`	Tab	M15	P0	All
S-36	Action-health analytics	`/o/:org/analytics/action-health`	Tab	M15	P0	All
S-37	Integrations	`/o/:org/settings/integrations`	Page	M05	P0	Admin

Inv #	Screen	Route / surface	Type	Module	Priority	Roles
S-38	Integration details	`.../integrations/:id`	Page	M05	P0	Admin
S-39	Source allowlists	`.../integrations/:id/allowlist`	Tab	M05	P0	Admin
S-40	Service catalog	`/o/:org/settings/services`	Page	M06	P0	Admin write, all read
S-41	Add/edit service	Modal	Modal	M06	P0	Admin
S-42	Organization settings	`/o/:org/settings/general`	Page	M02	P0	Admin
S-43	Members	`/o/:org/settings/members`	Page	M02	P0	Admin
S-44	Invitations	Members page tab	Tab	M02	P0	Admin
S-45	AI model settings	`/o/:org/settings/ai-model`	Page	M20	P0	Admin
S-46	Notification center	`/o/:org/notifications`	Page + popover	M19	P0	All
S-47	Notification settings	`/o/:org/account/notifications`	Page	M19	P0	All
S-48	Audit log	`/o/:org/settings/audit-log`	Page	M18	P0	Admin
S-49	Audit verification	Audit log action	Panel	M18	P0	Admin
S-50	Export options	Postmortem toolbar	Modal	M17	P0	All
S-51	Export progress/download	Export modal step	Modal	M17	P0	All
S-52	Global search	Command palette + `/o/:org/search`	Modal/Page	M16	P1	All
S-53	Saved filters	Filter bar menu	Popover	M16	P1	All
S-54	Bulk export	Library toolbar	Modal	M17	P1	All
S-55	Recurring causes analytics	Analytics tab	Tab	M15	P1	All
S-56	Postmortem timeliness analytics	Analytics tab	Tab	M15	P1	All

Inv #	Screen	Route / surface	Type	Module	Priority	Roles
S-57	Two-factor authentication	Profile section	Section	M01	P1	Auth
S-58	Reviewer assignment	Review workspace	Popover	M12	P1	Admin, Member
S-59	Suggestion mode	Editor mode	Mode	M10	P1	Admin, Member
S-60	GitHub issue linking	Action item drawer	Section	M14	P1	Admin, Member
S-61	Teams management	Settings	Page	M02	P1	Admin
S-62	Repeat-incident detection	Incident overview card	Section	M15	Later	Admin, Member
S-63	Repeat-incident analytics	Analytics tab	Tab	M15	Later	All
S-64	Audit-ready reports	Export modal option	Modal	M17	Later	All
S-65	Organization data archive	Settings general	Section	M17	P1	Admin
S-66	Shareable postmortem links	Export/share menu	Modal	M17	P2	All
S-67	Real-time collaborative editor	Editor	Mode	M10	Later	Admin, Member
S-68	Offline editing	Editor	Mode	M10	Later	Admin, Member
S-69	Voice dictation	Editor toolbar	Control	M10	Later	Admin, Member

Build rule: P0 rows are the v1 build list. P1 rows are built only if the schedule allows and only after the P0 acceptance criteria pass. "Later" rows get no UI in v1 except the reserved hooks listed in the module chapters (for example, a disabled placeholder is **not** added; code structure just must not block them).

PART 5. CROSS-CUTTING BEHAVIOR (EVERY SCREEN INHERITS THIS)

5.1 App shell



Control	Action	Result
Retrace wordmark	Click	Navigate `/o/:org`
Sidebar item	Click	Navigate; active item highlighted with accent
Sidebar collapse	Click	Toggle 240px / 64px rail; persisted in `localStorage`
OrgSwitcher	Click, choose org	Switch tenant context (M02), reload dashboard
OrgSwitcher "Create organization"	Click	Navigate `/orgs/new`
Notification bell	Click	Popover of latest notifications with "View all" to S-46; badge shows unread count
User menu	Click	Profile, Notification settings, Sign out, Sign out of all devices (GAP-06)
Breadcrumbs	Click segment	Navigate up

Services appears under Admin settings for write and also needs a read-only entry for Members: **Design choice:** show "Services" in the main (non-admin) nav group for everyone, with write controls gated by `services.manage`.

5.2 Mandatory states on every data view

Every list, table, chart and panel implements all four:

State	Rule
Loading	Skeletons matching the final layout. No full-page spinners except initial session restore.
Empty	EmptyState with one-sentence explanation and the primary next action button (for example "Create incident").
Error	ErrorState with a message from the error map, a Retry button, and "Copy reference" for the correlation id.
Permission denied	PermissionDenied panel, never a blank screen.

5.3 Forms

- Validation with Zod, errors shown inline under the field on blur and on submit; server 422 `details` map onto fields.
- Primary button shows a spinner and is disabled while submitting; double-submit is impossible.
- Success feedback: toast plus navigation or inline update.
- Unsaved changes in a modal or page prompt a confirm before closing.
- Secrets use write-only inputs (see 5.9).

5.4 Destructive actions

Always through ConfirmDestructiveDialog. Soft-deleted resources disappear from lists after success (response shape is GAP-07: handle both 204 and a resource body). Incident and organization deletion use type-to-confirm.

5.5 Asynchronous operations

Every 202 command follows this pattern:

1. Button shows pending; the response returns `job_id`.
2. A JobProgress component appears in place (not a separate page unless the module says so).
3. On `succeeded`: invalidate relevant queries, toast, move the user to the result.
4. On `failed`: show user-safe error, a Retry button when retryable, preserve prior data.

5. Leaving the page does not cancel the job. Returning shows the current job state (the job id is kept in the resource, or the latest job is looked up by `entity_id`; see GAP-03).

5.6 Optimistic concurrency and stale data

All mutable important resources carry an ETag. On **412** `stale_version` show StaleVersionDialog:

- Title: "This changed while you were working."
- For the editor: options **Reload latest** and **Keep my copy** (copy local content to clipboard/scratch before reloading). Autosave stops until the user chooses.
- For forms and drawers: **Reload** discards local edits after confirming.

Never auto-retry a 412.

5.7 Rate limits and job errors

429 shows a toast with the wait time (from `Retry-After`) and disables the triggering control until it elapses. Failed jobs show the user-safe error and a `correlation_id` reference. Default-model daily limit reached (AI) shows a dedicated LimitReached message with a link to AI settings (Admin) or "Ask an admin to configure a model" (Member).

5.8 Session expiry

Silent refresh first. If refresh fails while editing, keep the editor content in memory, show "Session expired. Sign in again to save your work" in a non-dismissible banner, open a login dialog (not a full redirect) that on success replays the pending save. Full-page redirect only when no unsaved work exists.

5.9 Secrets

API keys and tokens are write-only. Input type is password-style with a "Replace" flow when a value exists. After save, show only the masked hint (for example last four characters). Never put secret values in query params, local storage, logs, error reports or toasts.

5.10 Timestamps and times

All stored times are UTC ISO 8601. Display in the organization's `timezone` setting when present (display-only per DB design), else browser local, always with a "UTC" toggle on evidence and timeline views. Use TimestampLabel for evidence and timeline; use plain relative time ("2h ago", with exact time in tooltip) for everything else.

5.11 Copy and tone

Neutral, plain language, no blame wording anywhere in the product UI. Error text says what happened and what to do next. No em-dashes in UI copy.

PART 6. MODULE CHAPTERS

Each chapter uses the same template: **Purpose, Screens, Layout, Interactions table, States, Rules and edge cases, API dependencies, Acceptance criteria.** Interaction rows are written `control | trigger | action | result`. Endpoint paths are relative to `/api/v1/orgs/{org_id}` unless they begin `/api/v1`.

M01. Authentication and account

Purpose: get a person into the product safely. **Phase:** 1. **Screens:** S-01 to S-07, S-57.

Layout

Centered card (max 400px) on `--bg`, Retrace wordmark above, tagline "Hindsight, without the bias." below the form. No sidebar.

Interactions

ID	Control	Trigger	Action / API	Result
M01-X01	Sign in form (email, password)	Submit	`POST /api/v1/auth/login`	Success: load `/me`; go to `next` or last org or `/orgs/new` if no memberships. Failure: inline "Email or password is incorrect". 429: lockout message with wait time.
M01-X02	"Continue with GitHub" / "Continue with Google"	Click	Browser navigates `GET /api/v1/auth/oauth/{provider}/start`	Provider flow, returns to S-06
M01-X03	OAuth callback page	Load	`GET /api/v1/auth/oauth/{provider}/callback` (completed by browser redirect) then refresh session	Success: same routing as M01-X01. Failure: error message and link back to login.
M01-X04	Sign up form (name, email, password)	Submit	`POST /api/v1/auth/register`	"Check your email" screen; no auto sign-in until verified
M01-X05	Verify email page	Load with token	`POST /api/v1/auth/verify-email`	Success: "Email verified" and button to sign in. Expired or invalid token: message plus "Resend" (GAP-05)
M01-X06	"Forgot password?" link	Click	Navigate S-04	Email form
M01-X07	Forgot password form	Submit	`POST /api/v1/auth/password-reset`	Always shows neutral "If an account exists, we sent instructions" (no account disclosure)
M01-X08	Reset password form (new password)	Submit with token	`POST /api/v1/auth/password-reset` (confirm step, body is GAP-05)	Success: go to login with toast
M01-X09	Profile page	Load	`GET /api/v1/me`	Shows name, email, active org, role, memberships
M01-X10	"Sign out"	Click	`POST /api/v1/auth/logout`	Clear state, go to `/login`
M01-X11	"Sign out of all devices"	Click, confirm	GAP-06 (no documented endpoint)	Hide the control until the endpoint exists
M01-X12	Two-factor setup (P1)	Click	GAP (not in inventory)	Not built in v1

States

Session restore shows a branded full-screen loader. Login errors never reveal which of email or password was wrong. Locked accounts show the 429/lockout message.

Rules and edge cases

- Signed-in users visiting `/login` are redirected to their last org.
- `next` parameter is validated as a same-origin relative path before use.
- Password field shows strength hint; actual rules come from the server (422 details).
- Password reset and verify tokens are read from the query string then removed from the URL (`history.replaceState`).

API dependencies

Auth endpoints and `GET /me` (Fixed inventory). Verify and reset body fields are open (API Contracts 4.1): **GAP-05**.

Acceptance criteria

- [ ] M01-AC01 Valid login lands in the correct org; invalid login shows a generic error.
- [ ] M01-AC02 OAuth round trip works for GitHub and Google and handles failure.
- [ ] M01-AC03 Refresh on page reload restores the session without showing login.
- [ ] M01-AC04 Concurrent 401s trigger exactly one refresh call.
- [ ] M01-AC05 Password reset never discloses whether an email exists.
- [ ] M01-AC06 Access token is never written to storage.

M02. Organizations, workspace and members

Purpose: create and manage the private workspace. **Phase:** 1. **Screens:** S-08, S-42, S-43, S-44, S-61 (P1).

Interactions

ID	Control	Trigger	Action / API	Result
M02-X01	Create organization form (name, slug)	Submit	`POST /api/v1/orgs` (Idempotency-Key)	Navigate to SettingUpScreen; creator becomes Admin
M02-X02	SettingUpScreen	Load	Poll `GET /orgs/{org_id}` for `status`	`provisioning`: friendly progress copy. `active`: go to dashboard. `failed`: error with Admin "Retry" (provisioning retry endpoint is GAP-10)
M02-X03	OrgSwitcher list	Choose org	Core token for that org (GAP-11), clear caches	Navigate `/o/{org}`
M02-X04	Organization settings form (name, stall days, escalation days, retention)	Save	`PATCH /orgs/{org_id}`, `PATCH /orgs/{org_id}/settings` with `If-Match`	Toast, update cache. Fields shown: `stall_days`, `escalation_days`, `pre_due_days`, `evidence_retention_days`, `timezone` (DB design org_settings)
M02-X05	Members table	Load	`GET /members`	Columns: name, email, role, status, joined. Row actions menu
M02-X06	Change role (row menu)	Select role	`PATCH /members/{id}`	Toast; if it would remove the last Admin, server returns an error shown inline. Self-demotion shows a warning
M02-X07	Remove member (row menu)	Confirm	`DELETE /members/{id}`	Row removed; copy states "Their authored content stays; access ends"
M02-X08	"Invite member" dialog (email, role)	Submit	`POST /invitations` (Idempotency-Key)	Toast "Invitation sent"; appears in Invitations tab
M02-X09	Invitations tab	Load	**GAP-08** (no documented list endpoint)	Build the tab shell; if no endpoint, show invites created this session and a note. Do not invent the route
M02-X10	Revoke invitation	Confirm	`DELETE /invitations` (identifier shape GAP-08)	Row removed
M02-X11	Accept invitation page	Load with token (signed in or sign in first)	`POST /invitations/accept`	Success: go to org. Expired or revoked: clear message and "Ask an admin to resend"
M02-X12	Enable optional roles toggle	Click	Later (`enabled_optional_roles`)	Not built in v1; hide
M02-X13	Delete organization (P1)	Type org name to confirm	GAP-09 (no documented route)	Hidden in v1 unless endpoint documented

ID	Control	Trigger	Action / API	Result
M02-X14	Teams (P1)		No endpoint in inventory	Not built

Rules and edge cases

- Role selector in v1 offers Admin and Member only.
- The last active Admin cannot be demoted or removed (control disabled with tooltip, plus server enforcement).
- Org deletion, when built, shows the exact consequences: all incidents, evidence, drafts, comments and keys are permanently removed.
- Settings number fields enforce `> 0` for stall and escalation days (DB check).

Acceptance criteria

- [ ] M02-AC01 New org shows SettingUpScreen until `active`, then lands on the dashboard.
- [ ] M02-AC02 Switching orgs clears the previous org's cached data and never shows stale tenant content.
- [ ] M02-AC03 Last-Admin protections are enforced in UI and handled when the server rejects.
- [ ] M02-AC04 Admin-only screens redirect Members to NotFound-safe PermissionDenied.

M03. App shell navigation and Dashboard

Purpose: orient the user and surface what needs attention. **Phase:** 1 (shell) and 3 (metrics). **Screen:** S-09.

Layout

A 12-column grid. Row 1: four StatCards (headline metrics). Row 2: "Recent incidents" (8 cols) and "Postmortems awaiting review" (4 cols). Row 3: "Stalled action items" (6 cols) and "Open and overdue action items" (6 cols).

Interactions

ID	Control	Action / API	Result
M03-X01	"New incident" primary button (`incident.create`)	Open S-11 modal	
M03-X02	Recent incidents row	Click	Navigate to incident overview
M03-X03	Awaiting review row	Click	Navigate to that postmortem tab
M03-X04	Stalled / overdue / open row	Click	Open action item drawer (S-31)
M03-X05	StatCards	Click	Navigate to the matching analytics tab with filters preset
M03-X06	"View all" links	Click	Incident list, library, action list

Data sources: incidents list (`GET /incidents`, page 1 sorted `-started_at`), postmortems awaiting review (incidents list filtered by postmortem status, see GAP-01), action items (`GET /action-items` filtered), analytics summaries (`GET /analytics/*`). Headline metrics default to the latest period of `analytics_summaries`.

States

Each card loads independently; one failing card shows its own ErrorState and does not break the page.

Acceptance criteria

- [ ] M03-AC01 Dashboard renders under 2 seconds for typical data (PRD) with skeletons then data.
- [ ] M03-AC02 Every card has loading, empty and error states.
- [ ] M03-AC03 Members never see Admin-only nav items.

M04. Incidents

Purpose: list, create, view and edit incidents; entry point of the core loop. **Phase:** 1. **Screens:** S-10, S-11, S-12.

Incident list (S-10)

Table with columns: Severity, Title, Type, Status, Services, Started, Postmortem status, Owner/author. Filter bar: severity, status, service, type, date range. Sort menu. Page pagination. Row click opens the incident.

Incident detail frame (S-12)

Header: title (editable inline for authors), SeverityBadge, StatusBadge, services chips, started/ended times, source (PagerDuty or manual). Below, tabs: **Overview, Evidence, Timeline, Postmortem, Actions, Activity**. A "next step" banner above the tabs guides the loop (table below).

Incident state	Banner copy and primary action
Resolved, no description	"Tell Retrace what happened." Button: Add description
Description saved, discovery running	JobProgress "Finding evidence"
Evidence ready, unconfirmed	"Review the evidence we found." Button: Review evidence
Timeline ready, no postmortem	"Ready to draft." Button: Generate postmortem
Generating	JobProgress "Drafting postmortem"
Postmortem in progress	"Continue editing." Button: Open postmortem
Closed	"Postmortem published."

Interactions

ID	Control	Trigger	Action / API	Result
M04-X01	Filter bar, sort	Change	`GET /incidents` with declared filters (URL state)	Table refetches; filters in URL
M04-X02	"New incident" button	Click	Open S-11	
M04-X03	Create incident modal: required **short description**; optional severity, type, services, start/end	Submit	`POST /incidents` (Idempotency-Key) then `POST /incidents/{id}/description`	Navigate to the Evidence tab; discovery job starts (INC-3: description is the only required input)
M04-X04	Description field (resolved PagerDuty incident)	Submit	`POST /incidents/{id}/description`	`202` job; banner changes
M04-X05	Edit incident (title, type, severity, services, times, responders)	Save	`PATCH /incidents/{id}` with `If-Match`	Toast; 412 handled
M04-X06	Incident source / lifecycle timestamps	View	From `GET /incidents/{id}`	Show detected, acknowledged, mitigated, resolved with TimestampLabel
M04-X07	"Open postmortem"	Click	Navigate postmortem tab	
M04-X08	Delete incident (menu, Admin)	Type title to confirm	GAP-09 (no documented DELETE)	Hidden until documented. Copy ready: "Deletes its evidence, drafts and comments"
M04-X09	PagerDuty duplicates merge (P1)		Not in inventory	Not built

Rules and edge cases

- Severity values `sev1` to `sev4`; type values `outage, performance, security, data, other`.
- Editing `resolved_at` earlier than `started_at` is blocked client-side (DB check) with an inline error.
- A manually created incident has no PagerDuty times: show "Approximate start and end" fields (evidence window input per System Design 20.1).
- Incident list for a service that is archived still shows its snapshot name.

Acceptance criteria

- [ ] M04-AC01 An author can go from "New incident" to the evidence preview with one text field filled.
- [ ] M04-AC02 Filters, sort and page are in the URL and survive reload.
- [ ] M04-AC03 The next-step banner reflects each incident state in the table above.
- [ ] M04-AC04 A stale incident edit shows the stale dialog.

M05. Integrations and source allowlists

Purpose: connect Slack, PagerDuty (and GitHub as stretch) and control exactly what the AI may read. **Phase:** 2. **Screens:** S-37, S-38, S-39.

Layout

Integrations page: three provider cards (Slack, PagerDuty, GitHub) each showing status badge (active, error, revoked), account name, last sync, last error. GitHub card is marked "Stretch" in v1 and hidden if the feature flag is off (**Design choice:** feature flag `VITE_FEATURE_GITHUB`).

Interactions

ID	Control	Trigger	Action / API	Result
M05-X01	"Connect Slack"	Click	`POST /integrations/slack/connect` then redirect to provider OAuth	Return to `GET /integrations/slack/callback`, show success and go to allowlist step
M05-X02	"Connect PagerDuty"	Click	`POST /integrations/pagerduty/connect`	Show connected state and the webhook setup instructions (webhook URL is `POST /webhooks/pagerduty/{token}`; display exactly what the API returns, nothing constructed client-side)
M05-X03	"Connect GitHub" (stretch)	Click	`POST /integrations/github/connect`	App installation flow; same pattern
M05-X04	Integration card	Click	Navigate details S-38	Shows status, scopes, auth mode, connected by/at, health, masked hint where supported
M05-X05	Allowlist tab	Load	`GET /integrations/{id}/allowlist`	Table of allowed channels or repos with name, private flag, added by
M05-X06	Add or remove allowlist entries (searchable multi-select of channels or repos the provider exposes)	Save	`PUT /integrations/{id}/allowlist` (full replace)	Toast. Private channels require an extra confirm dialog
M05-X07	"Disconnect"	Confirm	`DELETE /integrations/{id}`	Status revoked; copy: "Retrace stops reading immediately"
M05-X08	"Use your own Slack app" (rate-limit workaround)	Expand advanced section	GAP-13 (field names not in contract)	Not built until contract defines credential fields
M05-X09	Slack export upload fallback		GAP-13	Not built in v1 UI
M05-X10	Time-window configuration		Discovery window lives on the incident (M07), not here	No control on this screen

Rules and edge cases

- Allowlist default: public incident channels (System Design); the UI preselects none and recommends channels by name matching only if the API provides suggestions (otherwise no suggestions).

- Direct messages and private channels are off by default; adding a private channel shows an explicit warning.
- Hosted-model warning: any page where evidence flows to a model (AI settings) states plainly that evidence leaves the tenant when a hosted model is used.
- Integration `last_error` is user-safe text; show and offer Reconnect when status is `error`.

Acceptance criteria

- [ ] M05-AC01 Slack and PagerDuty can be connected, listed, and disconnected.
- [ ] M05-AC02 Allowlist changes persist and require confirmation for private channels.
- [ ] M05-AC03 No token, key or secret is ever shown; only status and masked hint.
- [ ] M05-AC04 Admin-only: Members hitting these routes see PermissionDenied.

M06. Service catalog

Purpose: link services to repos, channels and PagerDuty so discovery is accurate with no per-incident setup. **Phase:** 2. **Screens:** S-40, S-41.

Interactions

ID	Control	Trigger	Action / API	Result
M06-X01	Services table	Load	`GET /services`	Name, owning team, description, link counts, archived toggle
M06-X02	"Add service"	Click	Open S-41 modal	
M06-X03	Service form (name, owning team, description)	Save	`POST /services` or `PATCH /services/{id}`	Toast; table updates
M06-X04	Links editor (repos, channels, PagerDuty service, runbook URL)	Save	`PUT /services/{id}/links`	Only allowlisted sources are selectable (DB rule: links must exist in allowlist). Non-allowlisted source shows "Add to allowlist first" with a link to M05
M06-X05	Archive/delete service	Confirm	`DELETE /services/{id}`	Archived services stay for history but are not offered for new incidents
M06-X06	Suggested links (P1)		No contract	Not built

Rules

- Service name unique among non-archived services (inline 409/422 mapped).
- If a service has no links, show a hint: discovery will fall back to PagerDuty data and keyword search.

- Members see the table read-only.

Acceptance criteria

- [ ] M06-AC01 Admin can add, edit, link and archive services.
- [ ] M06-AC02 Link pickers only list allowlisted resources.
- [ ] M06-AC03 Members see read-only view with no write controls.

M07. Evidence discovery and preview

Purpose: show what Retrace found, let the author correct it quickly, and lock a snapshot for drafting.
Phase: 2. **Screens:** S-13, S-14.

Flow summary

Describe, discovery job runs (JobProgress), **Evidence preview** appears, author optionally includes/excludes/adds, **Confirm**, timeline builds.

Layout (Evidence tab)

- Top: window card ("Window: 13:32 to 15:20 UTC", Edit link). Window is derived automatically (60 min before first alert to 30 min after resolution), visible and editable.
- Left (7 cols): evidence list grouped by source (PagerDuty, Slack, GitHub, Manual), each row: SourceIcon, kind, text preview (redacted), actor, TimestampLabel, ConfidenceMeter with "Why this score?" popover (reasons from `score_reasons`), include toggle.
- Right (5 cols): selected item detail: full redacted text, original link, redaction count, state.
- Footer bar: counts ("42 included, 18 excluded"), **Confirm evidence** primary button.

Interactions

ID	Control	Trigger	Action / API	Result
M07-X01	Discovery progress	Automatic after description	JobProgress on the `discover_evidence` job	On success the preview loads
M07-X02	Evidence list	Load	`GET /incidents/{id}/evidence` (page pagination, filter by source/state)	Rows by relevance
M07-X03	Include/exclude toggle	Click	`PATCH /evidence/{id}`	Row updates; creates a new immutable evidence snapshot; any running AI job becomes stale; show inline notice "Evidence changed. Regenerate to use it." (API 9.3)
M07-X04	"Add evidence" (paste link or text)	Submit	`POST /incidents/{id}/evidence`	Added as manual. Links to non Slack/GitHub/PagerDuty hosts are stored as notes and **not fetched**; show that explanation in the dialog
M07-X05	"Edit window"	Save	Window change on the incident (field names GAP-01b) then re-run discovery	Re-runs discovery with a confirm
M07-X06	"Confirm evidence"	Click	`POST /incidents/{id}/evidence/confirm`	`202` job `build_timeline`; navigate to Timeline tab when done
M07-X07	"Open original"	Click	New tab to `url`	External link icon, `rel="noopener noreferrer"`
M07-X08	Filter: source, included/excluded, failed	Change	Query params	List filters
M07-X09	Failed collection list	View	`evidence` items with state `failed`	Show provider, reason, "Retry collection" (re-sync on demand is P1; button hidden in v1)
M07-X10	Excluded history	Toggle "Show excluded"	Same list, state filter	Excluded stay visible, greyed, with one-click re-include

Rules and edge cases

- The preview step is **optional and takes seconds**: Confirm is always enabled once discovery succeeds.

- Excluded evidence must never reach generation; the UI makes the snapshot change explicit.
- Evidence text is already redacted; never attempt to un-redact. Show redaction count as a small badge.
- If discovery returns zero items, EmptyState: "We found no evidence in the allowed sources." Actions: add evidence manually, widen window, review allowlist (link to M05).
- If a source rate-limited, `stats` shows partial results: show a visible partial-collection banner with "Some sources were rate limited" (never hide).

Acceptance criteria

- [ ] M07-AC01 Include/exclude updates immediately and shows the stale-generation notice.
- [ ] M07-AC02 Pasting an external link adds a note without any fetch.
- [ ] M07-AC03 Timestamps use TimestampLabel and never fake precision.
- [ ] M07-AC04 Partial or failed collection is visibly reported.

M08. Timeline

Purpose: a single ordered, source-linked chronology with honest uncertainty. **Phase:** 2. **Screens:** S-15, S-16.

Layout

Vertical timeline grouped by **phase** (Detection, Diagnosis, Mitigation, Recovery, Other) as collapsible sections. Each event row: time (TimestampLabel), SourceIcon(s), one-line summary, key-moment badge (first symptom, detection, escalation, mitigation, resolution), DiscrepancyFlag, "Manual" tag, hidden events dimmed. Clicking a row opens the event Panel (S-16) on the right.

Interactions

ID	Control	Trigger	Action / API	Result
M08-X01	Timeline list	Load / scroll	`GET /incidents/{id}/timeline` cursor + limit	Infinite load; stable under inserts
M08-X02	Filter by source, phase, key moment; "show hidden"	Change	Query params	Filtered list
M08-X03	Event Panel	Click row	Local from loaded data	Summary, detail, sources, linked evidence (via `timeline_event_evidence`), timestamps (all five fields), confidence, discrepancy note
M08-X04	"Open source"	Click	New tab to evidence URL	
M08-X05	"Add event"	Submit form (time, summary, detail, key moment, phase)	`POST /timeline-events/{id}` per inventory (path shape GAP-14)	Added with Manual tag; survives timeline rebuild
M08-X06	Edit event	Save	`PATCH /timeline-events/{id}` `If-Match`	Marked manual
M08-X07	Hide event	Toggle	`PATCH /timeline-events/{id}` (`is_hidden`)	Dimmed; excluded from default view
M08-X08	Delete event	Confirm	`DELETE /timeline-events/{id}`	Removed; confirm text explains citations may reference it
M08-X09	Mark key moment	Menu	`PATCH`	Badge changes (P1 for editing; automatic detection is P0)
M08-X10	Discrepancy flag	Hover / click	Local	Explains "Sources disagree on time by more than the configured gap" and shows both times

Rules and edge cases

- Order is server-determined (`sort_key`); the client never re-sorts.
- Hiding or deleting an event cited by a postmortem shows a warning listing affected sections (the citation will become unresolved; see M10 rule "unresolved citation").
- A date-only or timezone-unknown event shows its limitation, never an exact minute.
- After the timeline changes, the postmortem tab shows "Timeline changed since this draft" with a Regenerate suggestion.

Acceptance criteria

- [ ] M08-AC01 Events appear in server order with phases and key moments.
- [ ] M08-AC02 Discrepancies and timestamp uncertainty are visible, not hidden.
- [ ] M08-AC03 Manual events are tagged and survive a rebuild.
- [ ] M08-AC04 Cursor paging does not duplicate or skip when events are added.

M09. AI generation, progress and clarification

Purpose: start drafting, show honest progress, recover from failure, and collect clarifications. **Phase:** 2. **Screen:** S-18.

Interactions

ID	Control	Trigger	Action / API	Result
M09-X01	"Generate postmortem" (`draft.generate`)	Click	`POST /incidents/{id}/postmortem/generate` (Idempotency-Key)	`202`; GenerationProgress replaces the empty state
M09-X02	Usage guard	Click when limit reached	Server response (limit error)	LimitReached notice (see 5.7)
M09-X03	GenerationProgress	Automatic	Poll `GET /jobs/{job_id}`; later SSE	Shows queued/running/succeeded/failed/**partial** (partial is an AI-specific outcome). Shows model name and prompt version when available
M09-X04	Retry failed	Click	Re-issue generate with a new Idempotency-Key	Collected evidence is preserved (never re-collected)
M09-X05	Retry partial	Click	Same	Reuses completed stages server-side
M09-X06	Result summary	On success	Refetch postmortem	Shows: claims counts by kind, **limitations** list, validation status, **VerificationStatusBanner**
M09-X07	VerificationStatusBanner	Render	From run metadata	If semantic verification unavailable: amber banner "Semantic verification unavailable. Deterministic checks passed." Never shown as a pass
M09-X08	Clarification questions	Render	From generation result (field GAP-15)	Cards, each with why-it-matters, related event chips and four actions: **Provide context, Add evidence, Mark unresolved, Continue without answering**
M09-X09	"Provide context"	Submit text	GAP-15 (endpoint for answers not in inventory)	Stored as author-provided context; rendered with `author_context` styling
M09-X10	"Add evidence"	Click	Navigate to evidence tab add dialog (M07-X04)	New snapshot; dependent stages rerun

ID	Control	Trigger	Action / API	Result
M09-X11	Token usage and cost (P1)	View	From run metadata	Small meta line

Rules and edge cases

- Do not infer or display hidden AI stage names from a progress integer (API 6.2).
- While generation runs, editing is allowed only on a previously saved version; the new result arrives as a new version, never overwriting.
- If the evidence snapshot changes mid-run, show "This run is out of date" with Regenerate.
- Clarification is a pause for the **affected stage only**; the rest of the draft remains usable.

Acceptance criteria

- [ ] M09-AC01 Generation shows progress and ends in success, partial or failed with correct UI.
- [ ] M09-AC02 Retry does not require re-collecting evidence.
- [ ] M09-AC03 Verification-unavailable is always displayed distinctly from passed.
- [ ] M09-AC04 Clarification answers appear as author context, never as verified facts.

M10. Postmortem editor (core product screen)

Purpose: a calm, fast, evidence-linked writing workspace with built-in AI. **Phase:** 2. **Screens:** S-17, S-19, S-20, S-21, S-23, S-59 (P1), S-67 to S-69 (Later).

Layout



Left section nav (collapsible) lists the 11 sections and jumps to them. The right panel is a single tabbed sheet (Citations, Comments, Blameless, Suggestions) that slides over; the column never reflows. **Focus mode** hides nav, panel and toolbar (toggle with `Cmd/Ctrl+.`).

Document model

TipTap with ProseMirror JSON (Fixed). Custom nodes and marks:

- `citation` inline node: attrs `claimId`, `eventIds[]`, `kind`.
- `comment` mark: attr `threadId` (maps to `anchor_mark_id`).
- `claimKind` decoration/mark: drives the visual treatment in 2.2.
- Section headings are real heading nodes with a stable `sectionKey` attribute (`summary, impact, timeline, root_cause, contributing_factors, detection, response, went_well, went_poorly, lessons, action_items`).

Interactions (editing)

ID	Control	Trigger	Action / API	Result
M10-X01	Open postmortem	Load	`GET /incidents/{id}/postmortem`	Content, `revision`, ETag, status, lock state
M10-X02	Typing	Edit	Local immediately; debounced autosave `PUT /postmortems/{id}` with `If-Match` and base revision	SaveStatus Saving then Saved. Autosave updates only the working draft, never creates a named version
M10-X03	Save failed / offline	Error	Keep local, retry with backoff	SaveStatus "Failed (Retry)"; content never discarded
M10-X04	Stale save	412	StaleVersionDialog	Reload or keep copy; autosave paused (also covers a second browser tab, PRD ED-7)
M10-X05	"Save version"	Click, optional label	Command surface is GAP-16	Disabled with tooltip until the explicit-save command is defined; autosave remains
M10-X06	Section nav item	Click	Scroll to section	Active section highlighted
M10-X07	Citation chip	Click	Open Citations panel with the event	Panel shows the timeline event (summary, source, times, original link). Unresolvable ids show "Source event unavailable" in warning style
M10-X08	Insert citation	Select text, "Cite evidence"	Open event picker (search timeline)	Inserts chip; claim kind chosen from a small menu
M10-X09	Read-only state	Postmortem `approved` or `published`	Editor non-editable	Banner "Approved. Reopen to edit." with Reopen button (Admin, Member per state machine); toolbar shows lock icon
M10-X10	Slash commands (P1) / shortcuts (P1)	`/` or keys	Local	Insert blocks, trigger actions
M10-X11	Library (S-17)	Load	**GAP-01**: no list endpoint for postmortems in the inventory	Build the library as the incident list filtered to incidents that have a postmortem, using incident filters; show status per postmortem from the postmortem resource when opened. Revisit when an endpoint exists

Interactions (inline AI, S-21)

ID	Control	Trigger	Action / API	Result
M10-X20	AIActionMenu	Select text	Floating menu: Make blameless, Clearer, Shorter, Expand with evidence, Custom instruction	
M10-X21	Run action	Choose	`POST /postmortems/{id}/ai/edit` via SSE (path details GAP-03)	Stream arrives into a SuggestionDiff: old text struck through, new text highlighted
M10-X22	Accept	Click / `Enter`	Apply the change to the document locally	Triggers normal autosave. New facts added by AI arrive with citation chips
M10-X23	Reject	Click / `Esc`	Discard	Suggestion removed; nothing saved
M10-X24	Cancel streaming	Click Stop	Abort the request	Partial text shown as a suggestion the user may accept or discard
M10-X25	AI error	Failure	`error` SSE event	Inline message with Retry; document untouched
M10-X26	Regenerate section	Section menu	`POST /postmortems/{id}/sections/{key}/regenerate`	`202`; result arrives as suggestion/new version (never overwrites human edits). Optional guidance text is P1
M10-X27	Suggestions not saved			Suggestions live in client memory only until accepted (API 6.5)

Interactions (comments, S-23)

ID	Control	Trigger	Action / API	Result
M10-X30	"Comment" on selected text	Click	`POST /postmortems/{id}/comments` (creates thread anchored to mark id)	Highlight in text; thread opens in panel
M10-X31	Comment list	Open panel	`GET /postmortems/{id}/comments`	Threads grouped Open, Resolved, Orphaned
M10-X32	Reply	Submit	`POST /postmortems/{id}/comments` with thread	Appended
M10-X33	Resolve / reopen thread	Click	`PATCH /comments/{id}`	Status changes (reopen supported where API allows)
M10-X34	Anchor follows text	Edit around comment	Local marks	If anchored text deleted, thread becomes **orphaned** and is shown in an Orphaned group with quoted text
M10-X35	Viewer comment-only	Role	`postmortem.comment` only	Editing disabled, commenting enabled

Rules and edge cases

- **One author at a time** in v1. A second tab triggers the stale flow on save; no live presence UI is built. The editor is structured (extensions array, document serialization isolated) so Yjs collaboration extensions can be added later without a rewrite.
- Role rules apply live: nobody edits while status is Approved or Published.
- The editor never writes an AI-generated claim into the document without citations or an explicit hypothesis/open-question label.
- Claim-kind styling comes from the document attributes; manually typed text has no kind and shows as plain text.
- Large documents: set a client-side soft warning near the document size limit once the server value is known (DB open item 5).
- Exports (M17) must include citations in readable form; the editor view and export use the same claim data.

Acceptance criteria

- [ ] M10-AC01 Typing feels instant; autosave never blocks input.
- [ ] M10-AC02 A stale second-tab save is detected and nothing is silently overwritten.
- [ ] M10-AC03 Every citation chip opens its event; broken ones are flagged.
- [ ] M10-AC04 Inline AI shows a diff, and nothing changes until Accept.
- [ ] M10-AC05 Approved and Published documents are not editable until reopened.
- [ ] M10-AC06 Comments stay attached as text moves and become orphaned when text is removed.

- [ ] M10-AC07 Hypotheses, open questions and author context are visually distinct and labeled in text, not only by color.

M11. Blameless language check

Purpose: flag and rewrite blame, with the author in control. **Phase:** 2. **Screen:** S-22.

Layout

Right-panel tab. Header: last-checked revision, counts (open, accepted, edited, dismissed). List of findings; each card shows the **matched text** (highlighted in the document when selected), **category** (attribution, judgment, counterfactual, absolute; also personal blame, unsupported intent, team blame, negligence, minimizing as the AI pipeline expands), a short **reason**, and a **suggested rewrite** when available.

Interactions

ID	Control	Trigger	Action / API	Result
M11-X01	"Run blameless check"	Click	`POST /postmortems/{id}/blameless-check`	`202`; findings refresh when done
M11-X02	Auto-check	On Submit for review	Same endpoint	Submit waits for a completed check on the latest revision
M11-X03	Findings list	Open tab	`GET /postmortems/{id}/blameless-suggestions`	Open findings first; stale findings (older revision) greyed with "Text changed"
M11-X04	Select finding	Click	Local	Scrolls to and highlights matched text via stored `anchor`
M11-X05	Accept rewrite	Click	`PATCH /blameless-suggestions/{id}` (`accept`) and apply replacement locally	Text replaced; autosave; recorded in version history
M11-X06	Edit rewrite	Edit then Apply	`PATCH` (`edit`)	Custom replacement applied
M11-X07	Dismiss	Click	`PATCH` (`dismiss`)	Finding closed
M11-X08	Rule-only mode	LLM unavailable	Findings without rewrite	Show "Suggested rewrite unavailable" and keep Dismiss/Edit manual
M11-X09	Reviewer view	Reviewer opens	Read-only list	Shows check status and counts; no actions unless they can edit
M11-X10	Show responders by role (P1)	Toggle	Not in inventory	Not built

Rules and edge cases

- The rule check works with no LLM; the UI must never imply the check failed because rewrites are missing.
- Accurate factual accountability ("the on-call engineer ran the rollback") is not blame; do not nag. Dismissal is always available.
- Counts shown on the Submit button tooltip: "3 open findings. You can still submit; reviewers will see them."
- **Design choice:** Open findings do not block submit (PRD requires the check to have run, not to be clean).

Acceptance criteria

- [ ] M11-AC01 Check runs automatically before submit and results are visible to reviewers.

- [ ] M11-AC02 Accept, edit and dismiss each persist and update counts.
- [ ] M11-AC03 Findings for older revisions are marked stale, not silently removed.
- [ ] M11-AC04 Rule findings appear even when no rewrite is available.

M12. Review and approval

Purpose: move a postmortem from draft to published with explicit human decisions. **Phase:** 2. **Screens:** S-26, S-27, S-28, S-58 (P1).

Layout

ApprovalBar sits under the editor toolbar and changes with status and role. A **Review panel** tab shows the review checklist (P1), reviewer list (P1), blameless status, unresolved comment count and approval history.

Interactions (state machine SM-1 in Part 7)

ID	Control	Visible when	Action / API	Result
M12-X01	"Submit for review"	Draft or Changes requested, can edit	`POST /postmortems/{id}/submit`	Status In review; editor stays editable per PRD role rules only until approval; reviewers notified. Disabled with reason if blameless check not run on latest revision or generation running
M12-X02	"Request changes"	In review, Admin	Dialog with required note	`POST /postmortems/{id}/request-changes`
M12-X03	"Approve"	In review, Admin, not the author	Confirm dialog showing version, open findings, unresolved comments	`POST /postmortems/{id}/approve`
M12-X04	Author viewing own postmortem	In review	Approve disabled with tooltip "You cannot approve your own postmortem"	Server also enforces (409/403)
M12-X05	"Publish"	Approved, Admin	`POST /postmortems/{id}/publish`	Status Published; incident may close when no critical action items are open
M12-X06	"Reopen"	Approved or Published, Admin or Member	Confirm dialog explaining a snapshot is stored first	`POST /postmortems/{id}/reopen`
M12-X07	Approval history	Open panel	From postmortem approvals data (via postmortem/versions; exact read route GAP-17)	List: approver, role at time, decision, note, time, version
M12-X08	"Assign reviewers" (P1)		Not in inventory	Not built
M12-X09	Review checklist (P1)		Not in inventory	Local-only UI is not allowed; skip until defined
M12-X10	Stale approval	Latest version changed after review	Server 409	Dialog: "The version changed. Review the latest version first." Refetch

Rules and edge cases

- Approver is Admin in v1 (Manager later). Button availability uses the `postmortem.approve` capability **and** the "not author" guard.
- Approve is tied to the **latest** version; the confirm dialog names the version label.

- Unresolved comments and open blameless findings are **warnings, not blockers**, in the approve dialog (Design choice).
- After Approve, show "Read-only. Reopen to make changes."

Acceptance criteria

- [ ] M12-AC01 State transitions and guards match SM-1 exactly.
- [ ] M12-AC02 Authors cannot approve their own postmortem in UI and server.
- [ ] M12-AC03 Reopen stores a snapshot and returns to In review.
- [ ] M12-AC04 Approval history shows approver, time and version.

M13. Versions

Purpose: a trustworthy, immutable history with compare and restore. **Phase:** 2. **Screens:** S-24, S-25.

Layout

Version history: vertical list (newest first) showing version number, label, trigger (generation, submit, request changes, approve, publish, reopen, restore, manual), creator, time, model and prompt version (if AI), blameless open count. Rolling queue: only the latest 15 named versions are active; a note says so.

Interactions

ID	Control	Trigger	Action / API	Result
M13-X01	Versions list	Load	`GET /postmortems/{id}/versions`	Newest first
M13-X02	Select version	Click	Local	Read-only preview of that snapshot
M13-X03	"Compare" (pick two)	Click	Client-side diff of the two snapshots' `content_text` / JSON	S-25: side by side or inline diff with additions/removals highlighted; citation changes shown
M13-X04	"Restore this version"	Confirm	**GAP-16** (no restore command in the inventory)	Hidden or disabled until defined. Restore must create a **new** version, never mutate history
M13-X05	Version provenance popover	Hover	From version fields	Trigger, label, creator, model, prompt version, blameless count
M13-X06	Rolling-queue note	Always	Static copy	"Older versions leave this list under the retention policy" (mechanics GAP-16)

Rules

Versions are immutable and read-only everywhere. Autosave drafts are not versions and never appear here.

Acceptance criteria

- [ ] M13-AC01 Versions list shows provenance for each entry.
- [ ] M13-AC02 Compare highlights textual and citation differences.
- [ ] M13-AC03 No UI path edits or deletes a named version.

M14. Action items and reminders

Purpose: make prevention work get done. **Phase:** 3. **Screens:** S-29 to S-32, S-60 (P1).

Layout

Action item page: tab bar (Board-like **List**, **Stalled**, **Overdue**, **Nearly due**) over a filterable table. Columns: Title, Priority, Owner, Due, Status, Stalled indicator, Linked incident. Row click opens the **Drawer** (S-31) with details, timeline of changes, verification block.

Interactions

ID	Control	Trigger	Action / API	Result
M14-X01	List with filters (owner, service, priority, status, stalled)	Load/Change	`GET /action-items`	URL-state filters
M14-X02	"New action item"	Click	Modal (S-30)	
M14-X03	Create form: title (required), description, owner, due date, priority, success criterion	Submit	`POST /action-items` (Idempotency-Key)	Created as confirmed only when owner and due date are both set (`confirmed_at` rule)
M14-X04	AI-proposed items (from postmortem Action items section)	View	From postmortem/draft proposals	Shown as **Suggested** with ai confidence; no owner or due date; clearly labeled "Not assigned"
M14-X05	Accept suggestion	Click	Opens create form prefilled; owner and due date must be chosen by a person	`POST /action-items`
M14-X06	Dismiss suggestion	Click	Local or `PATCH` drop (reason)	Item removed from suggestions
M14-X07	Change status (Open, In progress, Blocked, Done, Dropped)	Menu	`PATCH /action-items/{id}` `If-Match`	Dropped requires a reason (dialog). Done opens verification step
M14-X08	Edit fields	Save	`PATCH /action-items/{id}`	Toast; 412 handled
M14-X09	Verification (Done)	Submit note and/or link	`POST /action-items/{id}/verify`	Shows verification block; note encouraged, not required
M14-X10	Stalled indicator	Render	From `stalled_since`	Warning badge with "Stalled since X" and cause (overdue or no update for N days)
M14-X11	Change history	Open drawer	From item events	Created, status, owner, due, verified, confirmed entries
M14-X12	Linked incident / postmortem	Click	Navigate	
M14-X13	Create GitHub issue (P1) / Slack reminders (P1)		Not in inventory	Not built

ID	Control	Trigger	Action / API	Result
M14-X14	Reminder visibility	Notification center	Notifications (M19)	Owner sees pre-due, stalled, escalated notices

Rules and edge cases

- Stalled is **computed**, not a status. Never show it as a selectable status.
- Owner picker lists active org members (display names from `/members`).
- Dropped requires a non-empty reason; Done does not require a note.
- Items on a deleted incident disappear on refresh.

Acceptance criteria

- [ ] M14-AC01 No AI-proposed item is tracked or assigned without a person confirming owner and due date.
- [ ] M14-AC02 Dropped requires a reason; stalled displays correctly and is filterable.
- [ ] M14-AC03 Filters by owner, service, priority, status work and persist in the URL.

M15. Analytics

Purpose: show recovery performance, volume and action health. **Phase:** 3 (P0), later for repeat analytics. **Screens:** S-33 to S-36, S-55, S-56, S-62, S-63.

Layout

Analytics page with tabs: **Overview, Recovery time, Incident volume, Action health** (P1 tabs: Recurring causes, Timeliness; Later: Repeat incidents). A shared filter bar: period (week/month, range), service, severity. Charts use Recharts with tokens; every chart has a data table toggle and text summary for accessibility.

Interactions

ID	Control	Trigger	Action / API	Result
M15-X01	Recovery time tab	Load	`GET /analytics/recovery-times`	Cards: time to detect, acknowledge, mitigate, resolve (mean). Trend line chart by period
M15-X02	Incident volume tab	Load	`GET /analytics/incident-volume`	Stacked bars by severity or type, group-by selector (service, severity, type)
M15-X03	Action health tab	Load	`GET /analytics/action-health`	Completion rate, average time to close, stalled count
M15-X04	Filters	Change	Query params	Charts update, URL state
M15-X05	Chart point click	Click	Navigate to incident list filtered by the clicked bucket	
M15-X06	Empty analytics	No data	EmptyState	"Analytics appear after your first resolved incidents"
M15-X07	Repeat incident UI (Later)		Reserved	Incident overview will get a "Possible repeat" card with Confirm/Dismiss; no v1 UI

Rules

Durations display as human units ("12m 30s"), stored as seconds. Null means "No data", never zero. Demo-tenant data shows a small "Demo data" badge.

Acceptance criteria

- [ ] M15-AC01 The three P0 analytics views render from their endpoints with loading, empty and error states.
- [ ] M15-AC02 Filters by service, severity and period work and persist in the URL.
- [ ] M15-AC03 Missing values show "No data", not 0.

M16. Search and filtering

Purpose: find things fast. **Phase:** P1 (global search, saved filters); filters per list are P0 in their modules. **Screens:** S-52, S-53.

- Per-list filters and sorts are implemented inside their modules (M04, M14, M07, M08).
- **Global search (P1):** command palette (`Cmd/Ctrl+K`). **GAP-18:** no search endpoint is documented (PRD DSH-3 mentions Postgres full-text search). Until defined, the palette only searches navigation items and recently viewed incidents from local cache.

- **Saved filters (P1):** stored per user; **GAP-18** (no endpoint). Not built in v1.

Acceptance criteria

- [ ] M16-AC01 Palette opens with a shortcut and navigates to pages; it does not claim to search content until an endpoint exists.

M17. Exports and reports

Purpose: get postmortems out as PDF and Markdown with citations intact. **Phase:** 2 (single export), P1 and Later for bulk and reports. **Screens:** S-50, S-51, S-54, S-64, S-65, S-66.

Interactions

ID	Control	Trigger	Action / API	Result
M17-X01	"Export" button (postmortem toolbar)	Click	Open modal S-50	Choose format: PDF or Markdown. Shows "Includes timeline, source references, version label"
M17-X02	"Start export"	Click	`POST /postmortems/{id}/export` (format field per OpenAPI, Idempotency-Key)	`202`; modal moves to progress step
M17-X03	Export progress	Automatic	JobProgress	On success: "Download" button
M17-X04	"Download"	Click	`GET /exports/{id}/download` (signed link)	Browser downloads; link is short-lived, so request a fresh one on each click, never store it
M17-X05	Export failure	Error	User-safe message + Retry	
M17-X06	Export history (P1)		No list endpoint	Not built; modal only shows the current export
M17-X07	Bulk export / org archive / audit-ready report / share link		P1, Later, P2	Not built in v1. No placeholder UI

Rules

Which version is exported: default to the current version; display its label in the modal. Exports must preserve citations in readable, traceable form (AI-pipeline requirement).

Acceptance criteria

- [ ] M17-AC01 PDF and Markdown export both complete and download.
- [ ] M17-AC02 The download link is never cached or reused after expiry.

M18. Audit log

Purpose: a read-only, verifiable record. **Phase:** 3 (UI), 1 (backend log). **Screens:** S-48, S-49.

Layout

Admin-only table with cursor pagination: time, actor (name and role at the time), service, action, entity, summary. Filter by action, actor, entity, date. Row click opens a details drawer (details JSON shown as a safe key/value list; `details` never contains secrets).

Interactions

ID	Control	Trigger	Action / API	Result
M18-X01	Log table	Load/scroll	`GET /audit-log` cursor + limit	Infinite load
M18-X02	Filters	Change	Query params	URL state
M18-X03	Details drawer	Click row	Local	Full entry: action, entity, request id, hashes shown shortened with copy
M18-X04	"Verify integrity"	Click	`GET /audit-log/verify`	Panel S-49: result (OK or first mismatched entry) with timestamp of verification and head hash
M18-X05	Retention controls (P1)		Org settings (M02)	Not here

Rules

No edit or delete controls exist anywhere. Copy states entries are append-only and hash-chained.

Acceptance criteria

- [ ] M18-AC01 Log loads with cursor pagination and filters.
- [ ] M18-AC02 Verify shows OK or the first failing entry clearly.
- [ ] M18-AC03 No mutation affordance exists on this screen.

M19. Notifications

Purpose: in-app awareness of reviews, approvals and action reminders. **Phase:** 3. **Screens:** S-46, S-47.

Interactions

ID	Control	Trigger	Action / API	Result
M19-X01	Bell badge	Load, poll	`GET /notifications` (unread count from list)	Count badge; poll interval is a frontend config constant (no push in v1)
M19-X02	Bell popover	Click	Same list (latest)	Latest notifications with relative times; "View all"
M19-X03	Notification center	Load	`GET /notifications` page pagination, unread filter	Full list with type icon, title, body, time
M19-X04	Open notification	Click	`PATCH /notifications/{id}` (mark read), then navigate to `link_path`	Navigates inside the app
M19-X05	"Mark all as read"	Click	PATCH each unread (no bulk endpoint)	Do in sequence with progress; hide if many (GAP-19)
M19-X06	Notification settings	Load	`GET /notification-settings`	Switches: email, in-app; per-type mute
M19-X07	Save settings	Toggle	`PATCH /notification-settings`	Immediate save with toast
M19-X08	Reminder timing / thresholds (P1)		Org settings (M02 fields)	Admin only

Notification types (from DB design)

`draft_ready, review_requested, approved, changes_requested, action_due, action_stalled, action_escalated`. Each type maps to an icon and a deep link target.

Acceptance criteria

- [ ] M19-AC01 New notifications appear within the polling interval and deep-link correctly.
- [ ] M19-AC02 Mark-read persists and the badge updates.
- [ ] M19-AC03 Preferences save and are respected in the UI list (muted types are not shown as toasts).

M20. AI model settings and usage

Purpose: let an Admin choose the model with clear privacy messaging. **Phase:** 2 (setting), 3 (usage). **Screen:** S-45.

Layout

Two cards. **Model:** radio between **Retrace default model** and **Custom model**. Custom shows Provider (select: azure*openai, openai, anthropic, openai*compatible), Model name, Endpoint URL, API key (write-only), optional max context tokens. **Usage:** daily calls and limit on the default model; message when reached.

Interactions

ID	Control	Trigger	Action / API	Result
M20-X01	Load settings	Load	`GET /ai-model`	Shows current mode, provider, model name, endpoint, **masked key hint**, last test result and time
M20-X02	Switch to Default	Select, Save	`PUT /ai-model`	Toast
M20-X03	Custom config	Fill, Save	`PUT /ai-model`; key sent once	Connection test runs on save (PRD AI-3); show result inline
M20-X04	"Test connection"	Click	`POST /ai-model/test`	Inline success or user-safe failure; does not reveal the key
M20-X05	"Replace key"	Click	Opens an empty write-only field	Old key is never displayed
M20-X06	Privacy notice	Always visible	Static, plain	"Evidence sent to a hosted model leaves your organization workspace." (PRD AI model configuration)
M20-X07	Usage meter	Load	Usage data endpoint is not in the inventory	**GAP-20**: show limit messaging only when a generation returns the limit error; no usage chart until defined

Rules

Provider list is fixed to DB design values. Using a custom key removes the default-model limit (provider limits still apply); show that sentence in the custom section.

Acceptance criteria

- [ ] M20-AC01 The API key is never visible after saving; only a masked hint.
- [ ] M20-AC02 Connection test result is shown; failure does not discard other fields.
- [ ] M20-AC03 The hosted-model privacy notice is always visible on this screen.

M21. Jobs and system feedback (cross-cutting)

Purpose: one consistent way to show background work and system problems.

ID	Behavior	Rule
M21-X01	`useJob(jobId)` hook	Polls `GET /jobs/{job_id}` (start 2s, back off to a cap, stop at terminal state). Upgrades to SSE when the AI/Core SSE contract is published (GAP-03), keeping polling as fallback
M21-X02	Job states	queued, running, succeeded, failed, cancelled, plus AI "partial"
M21-X03	Retry	Offered only where retry is meaningful (generation, export, discovery) and creates a new command with a new Idempotency-Key
M21-X04	Cancel	Only where the service exposes it (none generic in v1). No cancel button otherwise
M21-X05	Job list view	Not required. Jobs are shown in place. The "view queued/running/failed jobs" inventory items are satisfied by per-screen JobProgress
M21-X06	Error reference	Every job error and API error shows a copyable `correlation_id`
M21-X07	Global banners	Session expiring, offline, rate limited (429), degraded service

Acceptance criteria

- [ ] M21-AC01 Every 202 flow uses JobProgress and handles all terminal states.
- [ ] M21-AC02 Navigating away and back during a job restores its progress view.

PART 7. END-TO-END WORKFLOWS AND STATE MACHINES

WF-1. Onboarding to first incident

```
Sign up -> verify email -> sign in -> Create organization -> SettingUpScreen (poll)
  -> Dashboard (empty) -> Settings: AI model (default preselected, confirm)
  -> Integrations: connect Slack, PagerDuty (GitHub stretch)
  -> Allowlist: choose channels -> Services: add services + links
  -> Members: invite teammates -> Create incident
```

Empty-dashboard guide: until setup is complete the dashboard shows a checklist (Choose AI model, Connect Slack, Connect PagerDuty, Add a service, Invite a teammate). Each item deep-links to the right settings screen and checks off from real state (integration status, service count, member count), not local flags.

WF-2. Incident to approved postmortem (the core loop)

```
Incident (from PagerDuty or manual)
   -> [M04] Describe what happened (only required input)
   -> [M07] Discovery job -> Evidence preview -> (optional include/exclude/add) -> Conf
  irm
   -> [M08] Timeline built (phases, key moments, discrepancies)
   -> [M09] Generate -> progress -> result, limitations, clarifications
   -> [M10] Edit with citations, inline AI, comments
   -> [M11] Blameless check (auto before submit) -> accept/edit/dismiss
   -> [M12] Submit for review -> reviewers comment -> Admin approves or requests change
  s
   -> Publish -> [M17] Export -> [M14] Action items tracked
```

Step	Screen	Primary control	Next
1	Create incident modal	"Create and find evidence"	Evidence tab with JobProgress
2	Evidence tab	"Confirm evidence"	Timeline tab with JobProgress
3	Timeline tab	"Generate postmortem"	Postmortem tab with GenerationProgress
4	Postmortem tab	Edit, comment, check	"Submit for review"
5	Postmortem tab (review)	"Approve" (Admin)	Locked read-only
6	Postmortem tab	"Publish"	Export, track actions

WF-3. Changes requested loop

```
In review -> Admin "Request changes" (note required) -> author sees banner with note
  -> author edits (blameless re-check if text changed) -> "Submit for review" -> In rev
  iew
```

WF-4. Reopen after approval

```
Approved/Published -> "Reopen" -> confirm (snapshot stored) -> In review (editable)
  -> edit -> submit -> approve again (new version, new approval record)
```

WF-5. Evidence changes after generation

```
Author excludes/includes evidence -> new immutable evidence snapshot
  -> running AI job marked stale -> banner "Evidence changed. Regenerate."
  -> Generate again -> new version created (previous versions keep their original snaps
  hot reference)
```

WF-6. Action item lifecycle

```
AI suggests (Suggested, unassigned) -> person confirms owner + due date -> Open
  -> In progress / Blocked -> Done (+ optional verification) | Dropped (+ required reas
  on)
  Stalled (computed): not Done AND (past due OR no update for N days)
  -> reminders: pre-due (owner) -> stalled (owner) -> escalated (Admins) -> in-app + em
  ail
```

WF-7. Invitation and join

```
Admin invites email+role -> invitee opens link -> sign in or sign up (same email)
  -> accept -> lands in org -> org appears in switcher
  Expired/revoked -> clear message, "Ask an admin for a new invite"
```

SM-1. Postmortem state machine (Fixed: PRD 9, System Design 21, API 10.1)

From	To	Who	UI guard
Draft	In review	Admin, Member	Blameless check run on latest revision, no generation running
In review	Changes requested	Admin	Note required
In review	Approved	Admin, not author	Reviewing the latest version
Changes requested	In review	Admin, Member	A new version exists (editing creates it)
Approved	Published	Admin	None
Approved or Published	In review	Admin, Member	Snapshot stored; read-only until reopened

Editable only in Draft, In review (by roles), Changes requested. Locked in Approved and Published.

SM-2. Incident states

Open, Resolved, Postmortem in progress, Closed. Transitions are system-driven (PagerDuty event or manual, generation, publish with no open critical action items). The UI never exposes manual state-change buttons for these except through the described actions.

SM-3. Action item states

Open, In progress, Blocked, Done, Dropped. Any non-terminal to Dropped requires a reason. Stalled is a flag.

SM-4. Evidence item states

Collected, Included, Excluded, Failed. Excluded items remain in history and can be re-included.

SM-5. Job states

Queued, Running, Succeeded, Failed, Cancelled (AI additionally exposes partial).

7.1 "Which file opens which" (navigation map)

From	Control	To
Dashboard	Recent incident row	Incident overview
Dashboard	Awaiting review row	Postmortem tab
Dashboard	Action row	Action item drawer
Incident list	Row	Incident overview
Incident overview	Banner button	Evidence / Timeline / Postmortem tab by state
Evidence tab	Confirm	Timeline tab
Timeline tab	Generate	Postmortem tab
Timeline event panel	Linked evidence	Evidence tab item
Postmortem editor	Citation chip	Citation panel, then Timeline event panel
Postmortem editor	Versions	Version history
Version history	Compare	Version comparison
Postmortem toolbar	Export	Export modal
Postmortem editor	Action items section	Action item create form
Action item drawer	Linked incident	Incident overview
Notification	Click	`link_path` target
Settings integrations	Card	Integration details, then allowlist
Service catalog	Link editor	(missing allowlist item) Integrations allowlist

PART 8. BUILD PLAN AND AGENT TASK TEMPLATE

8.1 Build order (matches the 8-week plan; one coherent PR per row)

Order	Work	Modules	Done when
1	Tokens, theme, base components, AppShell, router, providers	Design system, M03 shell	Shell renders all nav items with role gating
2	API client (auth, ETag, idempotency, errors), query keys, generated types pipeline (`pnpm generate:api`)	Architecture	One generated type consumed by one handwritten call
3	Auth screens + session restore	M01	M01 acceptance criteria pass
4	Org creation, SettingUpScreen, switcher, members, invitations, org settings	M02	A new org reaches the dashboard and invites a member
5	Incident list, create, detail frame, next-step banner	M04	Description-only create reaches Evidence tab
6	Integrations + allowlists, service catalog	M05, M06	Slack and PagerDuty connect; services link
7	Evidence preview + timeline	M07, M08	Description to confirmed timeline
8	JobProgress/useJob + generation + AI settings	M09, M20, M21	Generation completes, retry works
9	Editor core: TipTap, autosave, ETag/stale, citations, section nav	M10	M10-AC01 to AC03, AC05
10	Inline AI, comments, blameless panel	M10, M11	M10-AC04, AC06; M11 criteria
11	Review/approval, versions, export	M12, M13, M17	Incident to approved, exported postmortem
12	Action items + notifications	M14, M19	Suggested to done lifecycle and reminders visible
13	Analytics + audit log	M15, M18	Three analytics views + verify
14	Dashboard completion, empty states, a11y pass, demo data, E2E	M03, all	Playwright core loop passes

If the schedule slips, cut in the order from the PRD: email delivery (backend), analytics, GitHub, extra export formats. **Never cut:** tenant isolation tests, stale-version handling, citation resolution, human-accept for AI.

8.2 Task template (use for every Cursor, Antigravity or Codex task)

```
TASK: <Mnn / S-nn / WF-n, one line>
  ALLOWED PATHS: web/src/features/<x>/**, web/src/components/** (only if promoting shar
  ed code)
  READ FIRST: this document (Part 0.4, Part 5, the module chapter), Repository Conventi
  ons Spec, API and Event Contracts Spec section(s) <..>
  CONTRACTS: endpoints used (from the chapter), generated types in web/src/generated/
  DO: implement the chapter's interactions Mnn-X.., states, and acceptance criteria Mnn
  -AC..
  DO NOT: invent endpoints or fields, edit generated files, add libraries not listed in
  3.1 without an ADR, hide a GAP by guessing
  TESTS: Vitest + RTL for behavior (states, permission gating, error mapping); Playwrig
  ht only for WF-n critical flows
  HANDOFF: PR summary listing GAPs hit, files changed, screenshots of each state (loadi
  ng, empty, error, permission-denied)
```

8.3 Definition of done (every module PR)

- [ ] All interaction rows in the chapter implemented or explicitly listed as GAP/not-built with the documented fallback.
- [ ] Loading, empty, error and permission-denied states exist and are tested.
- [ ] Every mutation handles 412, 422, 429 and network failure.
- [ ] Role gating uses `can(capability)`; backend failure paths handled.
- [ ] No secrets or tenant data in logs, toasts or URLs.
- [ ] Keyboard navigation and focus order verified for the core flow.
- [ ] Acceptance criteria checkboxes in the chapter all pass.
- [ ] Generated types current (`pnpm generate:api` produces no diff).
- [ ] Conventional Commit messages and PR description per Repository Conventions.

8.4 Testing focus (frontend)

Area	Test
Session	Single-flight refresh, expiry during edit, org switch cache clear
Concurrency	412 on editor autosave and forms, no auto-retry
Idempotency	Retry reuses key, new intent gets new key
Permissions	Admin vs Member rendering per capability, 403 handling
Editor	Autosave debounce, citation click, read-only locks, suggestion accept/reject, comment anchoring
AI states	partial, failed, verification unavailable, limit reached
Evidence	Include/exclude shows snapshot notice; external link is note-only
Cross-tenant	Switching orgs never leaks cached data
A11y	Axe checks on core screens
E2E (Playwright)	WF-2 happy path against mocked backend

PART 9. GAPS AND ASSUMPTIONS REGISTER

Do not resolve a GAP by guessing in code. Build the documented fallback, mark the control, and record it in the PR. Resolve gaps in the central `contracts/` folder or an ADR first.

ID	Gap	Source of gap	Documented fallback in this doc
GAP-01	No `GET /postmortems` list endpoint for the library; no incident filter for "postmortem status"; evidence window field names	API Contracts 4.1	Library = incident list filtered by existing incident filters; status read per postmortem on open. Window edited through incident `PATCH` once fields are defined
GAP-02	Exact request/response fields, validation and filters for every endpoint	API-01	Use generated types only; do not hand-write unknown fields
GAP-03	Public job resource mapping, SSE path, reconnect and payloads; how to find the latest job for a resource	API-03	Poll `GET /jobs/{job_id}`; store `job_id` from the 202 response in client state; SSE adapter built behind `useJob` later
GAP-04	Page-size defaults, rate-limit numbers, retry/backoff values	CFG-01	Single frontend config constants file, clearly marked provisional
GAP-05	Verify-email, resend-verification and password-reset confirm bodies	API Contracts 4.1	Build screens; hide "Resend" if no endpoint
GAP-06	"Sign out of all devices" endpoint	Not in inventory	Control hidden
GAP-07	Soft-delete success response (204 vs body) and restore endpoints	API 10.3	Handle both; no restore UI
GAP-08	List and revoke invitations (identifier shape)	Not in inventory	Invitations tab shell with session-local list
GAP-09	Delete incident, delete organization, retry provisioning endpoints	Not in inventory	Controls hidden
GAP-10	Provisioning retry endpoint	System Design 4.5	SettingUpScreen shows error; Admin retry hidden
GAP-11	Exact call to switch the active organization token	API 3.2	Wrap in `switchOrg()` in `api/`; implement when documented
GAP-12	Notification settings route placement (`settings/*` Admin vs per-user)	System Design vs this doc	Placed under `account/`
GAP-13	Credential fields for org-owned Slack app; Slack export upload	System Design 15	Not built
GAP-14	Timeline event create path (inventory lists `POST, PATCH, DELETE /timeline-events/{id}`)	System Design 15	Follow OpenAPI once published; build the form now
GAP-15	Where clarification questions arrive and the endpoint to submit answers	AI Pipeline 10	Render from generation result if provided; "Provide context" disabled until endpoint exists

ID	Gap	Source of gap	Documented fallback in this doc
GAP-16	Explicit save command, restore command, rolling-queue archive mechanics	PM-01	"Save version" and "Restore" disabled with tooltip
GAP-17	Read route for approval history	Not in inventory	Derive from versions provenance and approve/decline entries when available
GAP-18	Search endpoint and saved filters	PRD DSH-3, DSH-4	Command palette navigates only
GAP-19	Bulk "mark all read"	Not in inventory	Sequential PATCH or hide
GAP-20	AI usage and limit read endpoint	Not in inventory	Show limit messaging only from error responses

Assumptions made in this document (all Design choices, change freely)

1. Dark theme only in v1; tokens are CSS variables so a light theme is additive.
2. Accent color `#7C8CFF`; fonts Inter, Source Serif 4, JetBrains Mono, self-hosted.
3. Hybrid layout: dense dashboard shell, calm editorial editor (max 720px column, right slide-over panel).
4. Open blameless findings and unresolved comments are warnings, not blockers, on submit and approve.
5. Services entry is visible to all members; write controls are Admin-only.
6. Notification settings live under `account/`, not `settings/`.
7. Sidebar collapse state is stored in `localStorage` (non-sensitive UI preference only).
8. GitHub UI is behind a feature flag in v1 (stretch).
9. Polling intervals (jobs, notifications) are frontend constants until contracts define better.

APPENDIX A. ENDPOINT TO SCREEN MAP

Prefix tenant paths with `/api/v1/orgs/{org_id}`. Auth and org creation use `/api/v1`.

Endpoint	Module	Screens
`POST /auth/register, /login, /logout, /refresh, /verify-email, /password-reset`, `GET /auth/oauth/{provider}/start, /callback`, `GET /me`	M01	S-01 to S-07
`POST /orgs`; `GET, PATCH /orgs/{org_id}`; `GET, PATCH /settings`	M02	S-08, S-42
`GET /members`; `PATCH, DELETE /members/{id}`; `POST, DELETE /invitations`; `POST /invitations/accept`	M02	S-43, S-44
`GET, POST /incidents`; `GET, PATCH /incidents/{id}`; `POST /incidents/{id}/description`	M04	S-10 to S-12
`GET /integrations`; `POST /integrations/{provider}/connect`; `GET .../callback`; `DELETE /integrations/{id}`; `GET, PUT /integrations/{id}/allowlist`	M05	S-37 to S-39
`GET, POST, PATCH, DELETE /services`; `PUT /services/{id}/links`	M06	S-40, S-41
`GET /incidents/{id}/evidence`; `PATCH /evidence/{id}`; `POST /incidents/{id}/evidence`; `POST /incidents/{id}/evidence/confirm`	M07	S-13, S-14
`GET /incidents/{id}/timeline`; `POST, PATCH, DELETE /timeline-events/{id}`	M08	S-15, S-16
`POST /incidents/{id}/postmortem/generate`	M09	S-18
`GET /incidents/{id}/postmortem`; `PUT /postmortems/{id}`; `GET /postmortems/{id}/versions`	M10, M13	S-19, S-24, S-25
`GET, POST /postmortems/{id}/comments`; `PATCH /comments/{id}`	M10	S-23
`POST /postmortems/{id}/ai/edit` (SSE); `POST /postmortems/{id}/sections/{key}/regenerate`	M10	S-21
`POST /postmortems/{id}/blameless-check`; `GET /postmortems/{id}/blameless-suggestions`; `PATCH /blameless-suggestions/{id}`	M11	S-22
`POST /postmortems/{id}/submit, /approve, /request-changes, /publish, /reopen`	M12	S-26 to S-28
`GET, POST /action-items`; `PATCH /action-items/{id}`; `POST /action-items/{id}/verify`	M14	S-29 to S-32
`GET /analytics/recovery-times, /incident-volume, /action-health`	M15	S-33 to S-36
`POST /postmortems/{id}/export`; `GET /exports/{id}/download`	M17	S-50, S-51
`GET /audit-log`; `GET /audit-log/verify`	M18	S-48, S-49
`GET /notifications`; `PATCH /notifications/{id}`; `GET, PATCH /notification-settings`	M19	S-46, S-47
`GET, PUT /ai-model`; `POST /ai-model/test`	M20	S-45
`GET /jobs/{job_id}`	M21	All async screens

APPENDIX B. COPY AND MESSAGE CATALOG (SEED)

Situation	Text
Stale save	"This changed while you were working. Reload the latest version or keep a copy of your text."
Own approval blocked	"You cannot approve your own postmortem."
Blameless not run	"Run the blameless check before submitting."
Verification unavailable	"Semantic verification unavailable. Deterministic checks passed."
No evidence found	"We found no evidence in the allowed sources. Add evidence, widen the time window, or review the allowlist."
Hosted model notice	"Evidence sent to a hosted model leaves your organization workspace."
Hypothesis tag	"Hypothesis. Not confirmed."
Author context tag	"Author context. Not independently verified."
Locked document	"Approved. Reopen to make changes."
Rate limited	"Too many requests. Try again in {n} seconds."
Session expired while editing	"Session expired. Sign in again to save your work."
Last admin	"An organization must keep at least one Admin."
Private channel	"Private channels can contain sensitive conversations. Add this channel only if the team expects Retrace to read it."

End of document. Version 1.0, 10 November 2026. Next revision should resolve GAP-01, GAP-03, GAP-15 and GAP-16 first, since they block the editor, job UX and generation clarification flows.